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Dear Editor,

We are pleased to submit our manuscript, "**Survey-Weighted Ordinal Causal Discovery: Application to Veteran Social Needs, Mental Health, and Well-Being**," for consideration in the **Annals of Applied Statistics**.

The paper addresses a methodological gap at the intersection of causal discovery and survey statistics. Existing causal discovery algorithms assume i.i.d. data, but modern surveys use unequal selection probabilities; applying these algorithms naively to survey data yields graphs that reflect the sampling design rather than the population-level causal structure. To our knowledge, this problem has not been formally addressed in the causal discovery literature.

We introduce the **survey-weighted, covariate-adjusted ordinal Bayesian network (swa-oBn).** The method integrates Horvitz-Thompson weights and Kish's effective sample size into a pseudo-BIC scoring framework, accommodates exogenous covariate adjustment without expanding the search space, presents two scoring variants (single-BIC and split-BIC) with a regime-based selection rule for heterogeneous ordinal scales, and supports structural constraints from domain knowledge. We prove that the framework consistently recovers the true population-level causal graph under a joint superpopulation-and-design asymptotic framework.

We apply swa-oBn to national VA CAHPS survey data to map the causal architecture of social determinants of health among U.S. Veterans. The weighted analysis identifies unmet basic needs as the upstream causal hub, with discrimination and isolation serving as proximal mediators to mental health and well-being. An unweighted analysis of the same data places discrimination upstream of material deprivation — a qualitatively different and, under the stated assumptions, incorrect causal ordering. Complementary simulations confirm that the survey-weighting problem extends to other algorithm families (PC, FCI, LiNGAM).

The CAHPS dataset has appeared in three recent companion publications (Frank et al. 2025, JAMA Health Forum; Lamba et al. 2025, JAMA Network Open; Russell et al. 2026, JAMA Network Open), of which I am a co-author on the latter two. Those analyses used associational methods to characterize prevalence and risk-need concordance across demographic strata; the present paper develops new methodology and addresses different scientific questions about causal structure among social need domains. There is no analytic or textual overlap.

Regarding reproducibility: the raw CAHPS data cannot be made publicly available due to VHA privacy restrictions, but all analysis code and fully reproducible simulation studies will be released openly.

This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. Thank you for your time and consideration.

Soumik Purkayastha

Sincerely,
Soumik Purkayastha.